DESIGN AND DEVELOP AN MQTT – BASED REMOTE HVAC MONITORING SYSTEM USING ThingSpeak CLOUD

ASSIGNMENT – 14

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PROJECT OVERVIEW

The MQTT-Based Remote HVAC Monitoring System using ThingSpeak Cloud is an IoT-based monitoring and control system developed to remotely observe important HVAC environmental parameters such as temperature and humidity. The system uses a DHT22 sensor connected to an ESP8266 NodeMCU, which collects the sensor readings and processes them for transmission.

The ESP8266 uses Wi-Fi connectivity to communicate with the MQTT system. MQTT provides a lightweight and reliable method for transferring sensor information between the monitoring unit and the cloud environment. The collected data is uploaded to ThingSpeak Cloud, where it can be stored and displayed through graphical visualizations for real-time and historical analysis.

A 16×2 LCD display is included to provide local information about the measured temperature and humidity values. A relay module is used to control the HVAC equipment according to predefined operating conditions or remote commands.

The cloud platform enables users to access HVAC information remotely without being physically present near the equipment. The stored data can also be analyzed to identify environmental variations, monitor HVAC performance, and support better operating decisions.

Overall, the proposed system provides real-time monitoring, wireless communication, cloud data storage, visualization, and remote HVAC control, making HVAC management more convenient, efficient, and suitable for IoT-based smart building applications.

PROBLEM STATEMENT

HVAC systems are widely used in residential, commercial, and industrial environments to maintain comfortable and suitable indoor conditions. However, conventional HVAC systems often depend on local monitoring, making it difficult for users to continuously observe temperature, humidity, and system status from a remote location.

Manual monitoring can result in delayed identification of abnormal temperature or humidity conditions. It can also make it difficult to maintain suitable environmental conditions and respond quickly to changes in the HVAC operating environment.

Another major limitation is the lack of continuous cloud-based data storage. Without historical sensor data, users have limited ability to analyze environmental variations, identify operating patterns, and evaluate HVAC performance over time.

Therefore, there is a need for an IoT-based HVAC monitoring system that can:

- Continuously monitor temperature and humidity conditions.
- Provide real-time HVAC parameter information.
- Transmit sensor data wirelessly using Wi-Fi.
- Use MQTT for lightweight and efficient communication.
- Store sensor readings on ThingSpeak Cloud.
- Provide graphical visualization of real-time and historical data.
- Display current readings locally using a 16×2 LCD.

- Provide relay-based HVAC control when required.
- Allow users to monitor HVAC conditions remotely.
- Support analysis of collected data for improved HVAC operation and energy management.

The proposed MQTT-Based Remote HVAC Monitoring System using ThingSpeak Cloud addresses these limitations by integrating sensor monitoring, ESP8266-based processing, MQTT communication, cloud storage, data visualization, and relay-based control into a single IoT-enabled system.

PROBLEM SOLUTION

The proposed MQTT-Based Remote HVAC Monitoring System provides an IoT-based solution for monitoring and controlling HVAC conditions remotely. The system continuously measures temperature and humidity using a DHT22 sensor connected to an ESP8266 NodeMCU.

The ESP8266 processes the sensor readings and transmits the data through Wi-Fi using the MQTT protocol. The sensor information is uploaded to ThingSpeak Cloud, where it is stored and displayed through real-time graphs for easy monitoring and analysis.

A 16×2 LCD provides local visualization of the current temperature and humidity values. A relay module is used to control the HVAC system based on predefined temperature conditions or remotecontrol requirements.

The cloud-based system allows users to monitor HVAC parameters from any location with Internet access. Historical data stored in ThingSpeak can also be analyzed to identify environmental trends and improve HVAC operating efficiency.

KEY SOLUTION

- Provides continuous temperature and humidity monitoring.
- Enables remote HVAC monitoring through ThingSpeak Cloud.
- Uses MQTT for lightweight and efficient data communication.
- Stores sensor data for historical analysis.
- Provides real-time graphical visualization.
- Displays sensor readings locally using a 16×2 LCD.
- Enables HVAC ON/OFF control through a relay.
- Reduces dependence on manual monitoring.
- Helps identify abnormal environmental conditions quickly.
- Supports improved HVAC performance and energy management.

SYSTEM ARCHITECTURE

The MQTT-Based Remote HVAC Monitoring System consists of sensing, processing, communication, cloud monitoring, display, and control sections.

1.Sensing unit

The DHT22 sensor measures the temperature and humidity of the HVAC environment. These parameters are continuously provided to the ESP8266 NodeMCU for processing.

2. Processing Unit

The ESP8266 NodeMCU acts as the main controller. It reads the DHT22 sensor values, processes the readings, and prepares the data for transmission.

3.Communication Unit

The ESP8266 connects to the Internet through Wi-Fi. MQTT is used as the communication protocol because it provides lightweight and efficient transmission of sensor data.

4. Cloud Unit

ThingSpeak Cloud receives and stores the HVAC sensor data. The cloud platform provides graphical visualization of temperature and humidity readings and allows historical data to be analyzed.

5. Display Unit

A 16×2 LCD is connected to the ESP8266 to display the current temperature and humidity readings locally.

6. Control Uint

A relay module is connected to the ESP8266 for controlling the HVAC equipment. The relay can switch the HVAC system ON or OFF according to predefined conditions or control commands.

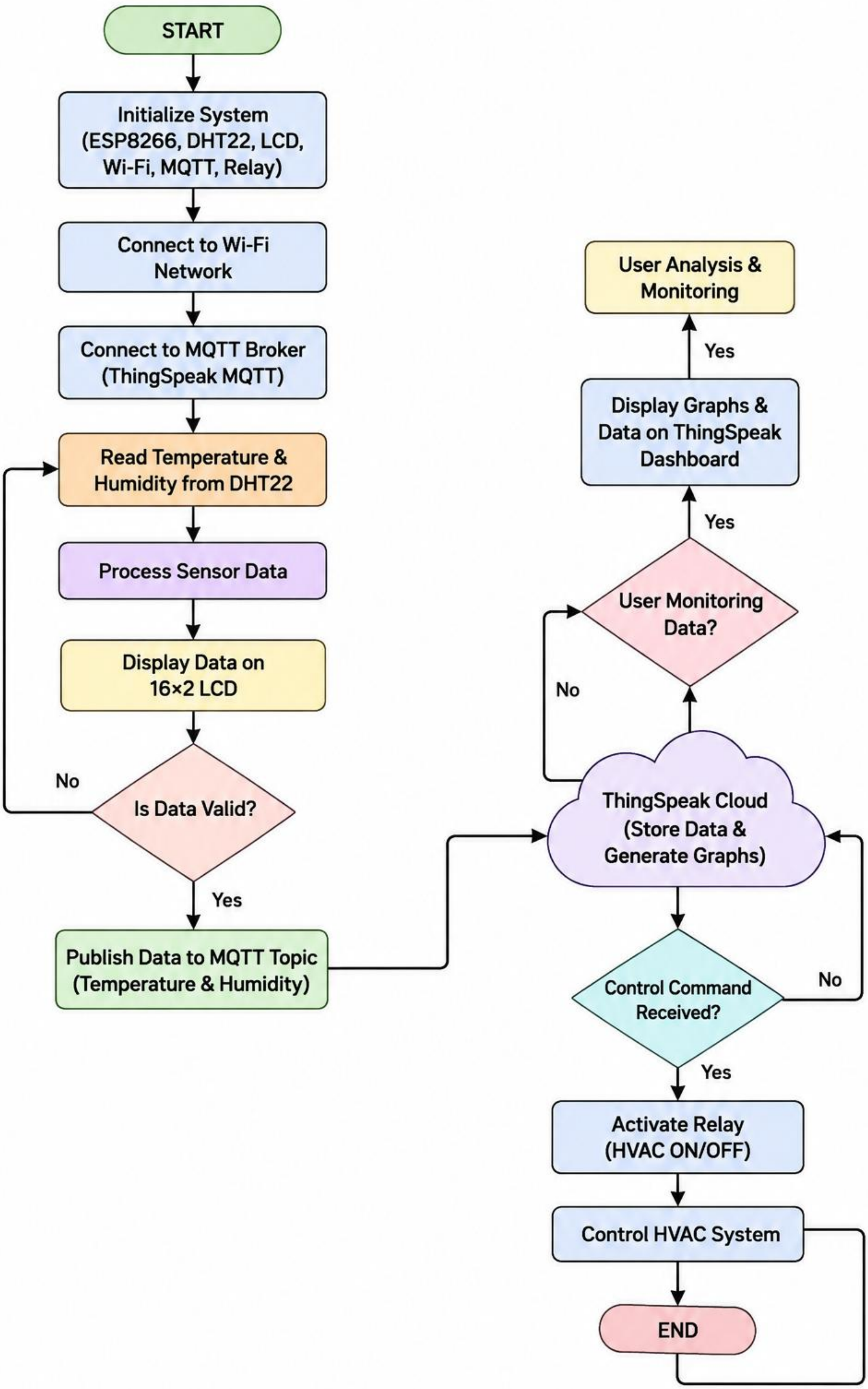
7. Remote Monitoring

The user can access the ThingSpeak dashboard remotely through an Internet-connected device. This allows HVAC conditions to be monitored without being physically present near the system.

Main Architecture

DHT22 Sensor → ESP8266 NodeMCU → Wi-Fi → MQTT → ThingSpeak Cloud

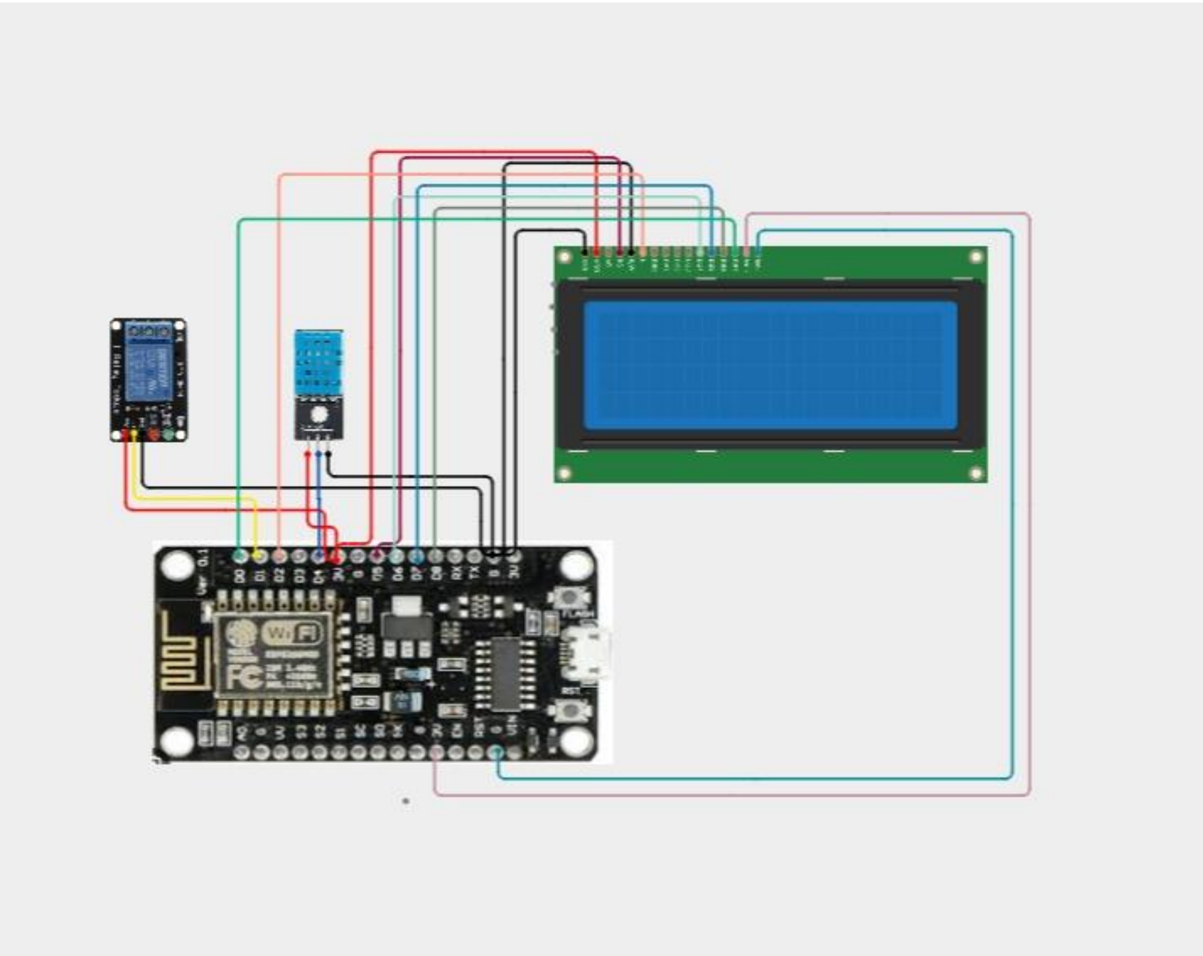
ESP8266 NodeMCU → 16×2 LCD
ESP8266 NodeMCU → Relay Module → HVAC Control
ThingSpeak Cloud → Remote User Monitoring


START
Initialize System (ESP8266, DHT22, LCD, Wi-Fi, MQTT, Relay)
Connect to Wi-Fi Network
Connect to MQTT Broker (ThingSpeak MQTT)
Read Temperature & Humidity from DHT22
Process Sensor Data
Display Data on 16×2 LCD
Is Data Valid?
No
Yes
Publish Data to MQTT Topic (Temperature & Humidity)
ThingSpeak Cloud (Store Data & Generate Graphs)
User Monitoring Data?
No
Yes
Display Graphs & Data on ThingSpeak Dashboard
Yes
User Analysis & Monitoring
Control Command Received?
No
Yes
Activate Relay (HVAC ON/OFF)
Control HVAC System
END

CIRCUIT TABLE

COMPONENT	PIN	ESP-8266 PIN
DHT22	VCC, DATA, GND	3.3V, D1, GND
Relay	Vcc, IN, GND	3.3V, D1, GND
16x2 LCD	RS, E, D4-D7	D5, D6, D2, D3, D7, D8
LCD	VSS, VDD, RW	GND, 3.3V, GND
LCD	A, K	3.3V, GND
Relay	COM, NO	HVAC Supply
ESP8266	USB	5V

CIRCUIT DESIGN



The circuit is designed around an ESP8266 NodeMCU, which acts as the central controller of the HVAC monitoring system. It receives data from the DHT22 sensor, processes the measured temperature and humidity values, and manages communication with the cloud platform.

The DHT22 sensor is connected to the NodeMCU through its digital data pin. It continuously measures the surrounding temperature and humidity and sends the readings to the controller. The 16×2 LCD is connected to the NodeMCU to display the measured values locally, allowing users to observe the current HVAC conditions without accessing the cloud.

A 1-channel relay module is connected to a digital output pin of the ESP8266. The relay provides electrical switching for the HVAC system and can turn the equipment ON or OFF according to the programmed conditions or received control commands.

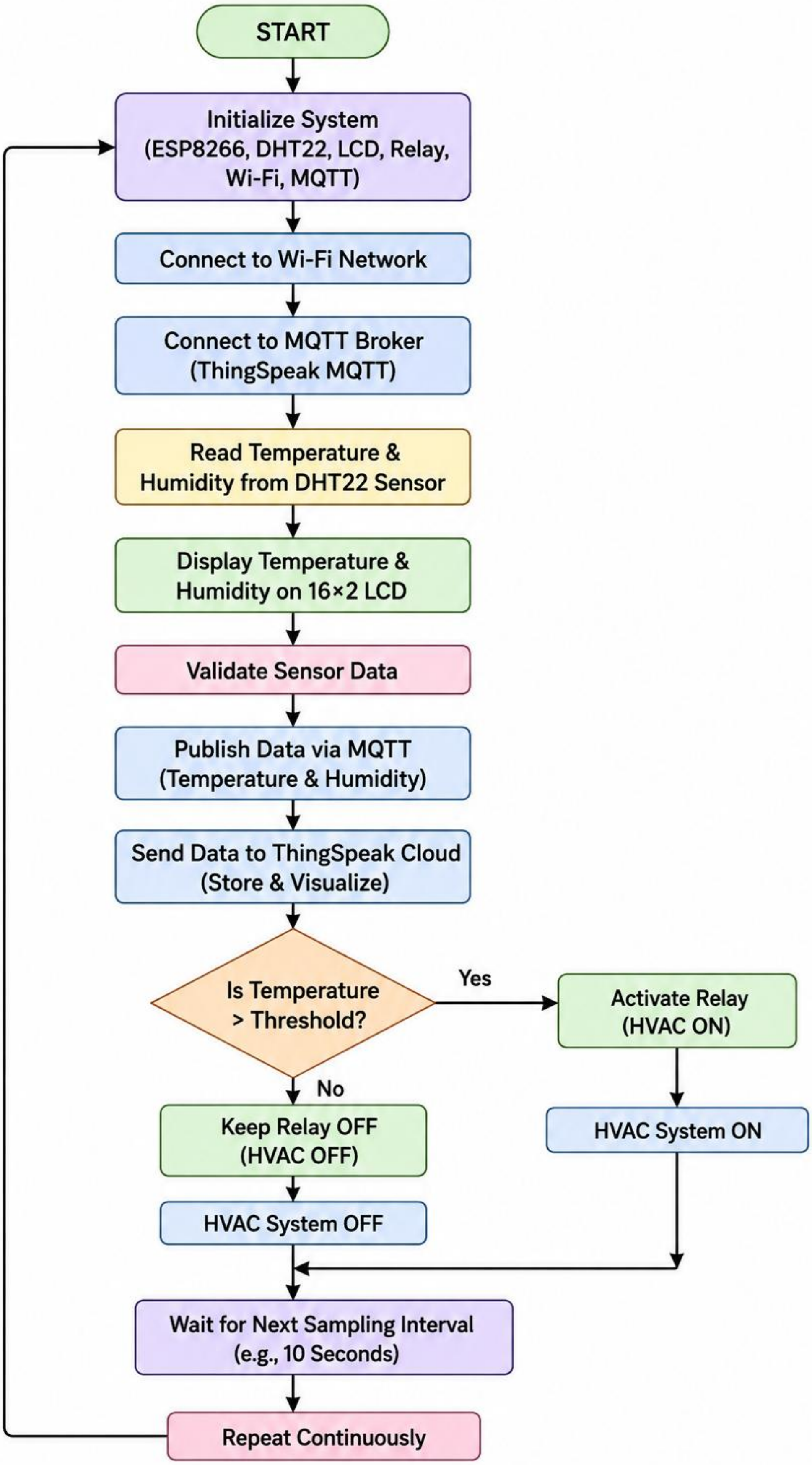
The components are assembled on a breadboard using jumper wires, with common power and ground connections provided to the sensor, display, and relay circuit. The ESP8266 is powered through its USB connection, while the connected components receive the required supply from the controller or appropriate power source.

For remote monitoring, the ESP8266 connects to a Wi-Fi network and communicates using the MQTT protocol. The sensor readings are transmitted through MQTT and made available to ThingSpeak Cloud for storage, visualization, and analysis. This allows users to monitor HVAC parameters remotely and observe changes in temperature and humidity over time.

The system therefore combines local sensing, LCD-based display, relay-based control, wireless MQTT communication, and cloud monitoring into a single compact HVAC monitoring and control system.

ALGORITHM

- Start the system.
- Initialize ESP8266, DHT22 sensor, 16×2 LCD, relay, Wi-Fi, and MQTT.
- Connect the ESP8266 to the Wi-Fi network.
- Establish the MQTT connection.
- Read temperature and humidity from the DHT22 sensor.
- Display the readings on the 16×2 LCD.
- Process and validate the sensor data.
- Publish the readings through MQTT.
- Send the data to ThingSpeak Cloud for storage and visualization.
- Compare the temperature with the predefined threshold.
- If the temperature exceeds the limit, activate the relay; otherwise, keep it OFF.
- Continue monitoring and repeat the process continuously.


START
Initialize System (ESP8266, DHT22, LCD, Relay, Wi-Fi, MQTT)
Connect to Wi-Fi Network
Connect to MQTT Broker (ThingSpeak MQTT)
Read Temperature & Humidity from DHT22 Sensor
Display Temperature & Humidity on 16×2 LCD
Validate Sensor Data
Publish Data via MQTT (Temperature & Humidity)
Send Data to ThingSpeak Cloud (Store & Visualize)
Is Temperature > Threshold?
Yes
Activate Relay (HVAC ON)
HVAC System ON
No
Keep Relay OFF (HVAC OFF)
HVAC System OFF
Wait for Next Sampling Interval (e.g., 10 Seconds)
Repeat Continuously

CODING

```
#include <ESP8266WiFi.h>
#include <PubSubClient.h>
#include <DHT.h>
#include <LiquidCrystal.h>

// ---------- Wi-Fi ----------
const char* ssid = "YOUR_WIFI_NAME";
const char* password = "YOUR_WIFI_PASSWORD";

// ---------- MQTT ----------
const char* mqtt_server = "mqtt3.thingspeak.com";
const int mqtt_port = 1883;

const char* mqtt_client_id = "YOUR_CLIENT_ID";
const char* mqtt_username  = "YOUR_MQTT_USERNAME";
const char* mqtt_password  = "YOUR_MQTT_PASSWORD";

const char* mqtt_topic = "channels/YOUR_CHANNEL_ID/publish";

// ---------- DHT22 ----------
#define DHTPIN D4
#define DHTTYPE DHT22

DHT dht(DHTPIN, DHTTYPE);

// ---------- Relay ----------
#define RELAY_PIN D1

// ---------- 16x2 LCD ----------
```

```
LiquidCrystal lcd(D5, D6, D2, D3, D7, D8);

WiFiClient espClient;
PubSubClient client(espClient);

float temperature;
float humidity;

const float TEMP_LIMIT = 30.0;

void setup() {
  Serial.begin(115200);

  dht.begin();

  pinMode(RELAY_PIN, OUTPUT);
  digitalWrite(RELAY_PIN, LOW);

  lcd.begin(16, 2);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("HVAC MONITOR");
  lcd.setCursor(0, 1);
  lcd.print("Starting...");

  WiFi.begin(ssid, password);

  Serial.print("Connecting WiFi");

  while (WiFi.status() != WL_CONNECTED) {
```

```
    delay(500);
    Serial.print(".");
  }

  Serial.println();
  Serial.println("WiFi Connected");
  Serial.println(WiFi.localIP());

  client.setServer(mqtt_server, mqtt_port);
}

void reconnectMQTT() {

  while (!client.connected()) {

    Serial.print("Connecting MQTT...");

    if (client.connect(
          mqtt_client_id,
          mqtt_username,
          mqtt_password)) {

      Serial.println("Connected!");

    } else {

      Serial.print("Failed, State = ");
      Serial.println(client.state());

      delay(5000);
```

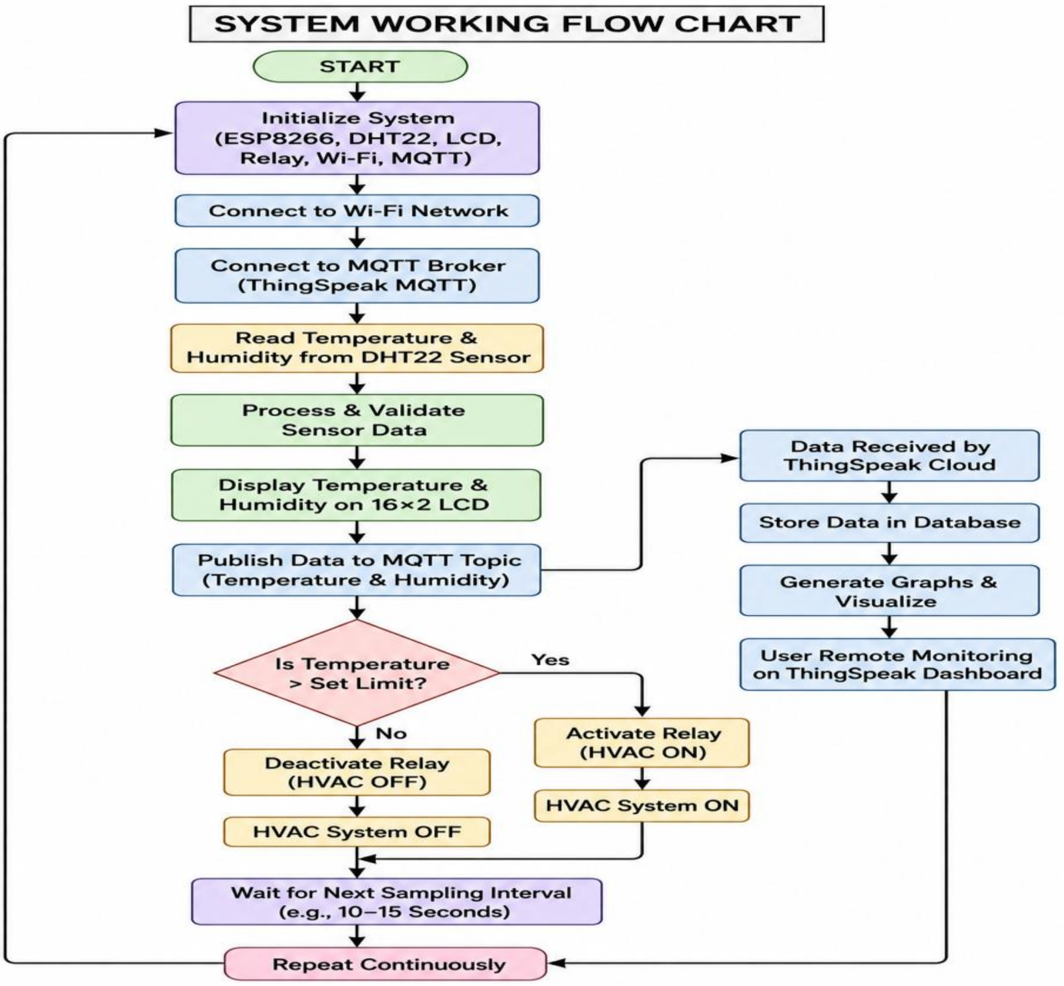
```
    }
  }
}

void loop() {

  if (!client.connected()) {
    reconnectMQTT();
  }

  client.loop();

  // Read sensor
  temperature = dht.readTemperature();
  humidity = dht.readHumidity();

  if (isnan(temperature) || isnan(humidity)) {

    Serial.println("DHT22 Sensor Error");

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Sensor Error");

    delay(2000);
    return;
  }

  // Display on LCD
  lcd.clear();
```

```
lcd.setCursor(0, 0);
lcd.print("Temp: ");
lcd.print(temperature, 1);
lcd.print((char)223);
lcd.print("C");

lcd.setCursor(0, 1);
lcd.print("Hum : ");
lcd.print(humidity, 1);
lcd.print("%");

// HVAC control
if (temperature > TEMP_LIMIT) {

  digitalWrite(RELAY_PIN, HIGH);

  Serial.println("HVAC: ON");

} else {

  digitalWrite(RELAY_PIN, LOW);

  Serial.println("HVAC: OFF");
}

// MQTT payload
String payload =
  "field1=" + String(temperature, 2) +
  "&field2=" + String(humidity, 2) +
```

```
    "&field3=" + String(digitalRead(RELAY_PIN));

  Serial.print("Publishing: ");
  Serial.println(payload);

  // Publish to ThingSpeak MQTT
  client.publish(mqtt_topic, payload.c_str());

  delay(15000);
}
```

SYSTEM WORKING

- The ESP8266 NodeMCU is powered on and initializes the DHT22 sensor, 16×2 LCD, relay module, Wi-Fi, and MQTT communication.
- The DHT22 sensor measures the surrounding temperature and humidity at regular intervals.
- The ESP8266 receives the sensor readings and processes the data to ensure that valid temperature and humidity values are obtained.
- The current temperature and humidity values are displayed on the 16×2 LCD for local monitoring.
- The ESP8266 connects to the Internet through Wi-Fi and establishes communication with the MQTT broker.
- The processed sensor data is published using the MQTT protocol. MQTT provides lightweight and efficient communication between the ESP8266 and the cloud system.
- The data is sent to ThingSpeak Cloud, where temperature and humidity values are stored and displayed using graphs. This allows users to monitor HVAC conditions remotely.
- The measured temperature is compared with a predefined temperature limit. If the temperature exceeds the set limit, the ESP8266 activates the relay to turn the HVAC system ON.
- When the temperature is within the desired range, the relay remains OFF or switches the HVAC system OFF according to the programmed control logic.
- The system repeats the sensing, processing, display, cloud transmission, and control operations continuously at the selected sampling interval.



BILL OF MATERIAL

COMPONENT	QUANTITY
ESP8255 NodeMCU	1
DHT22 Sensor	1
Relay Module	1
OLED/LCD Display	1
Breadboard	1
Jumper Wires	1 Set

PROTOTYPE IMPLEMENTATION

The prototype is implemented using an ESP8266 NodeMCU as the main processing and communication controller. A DHT22 sensor is connected to the NodeMCU to measure the surrounding temperature and humidity continuously.

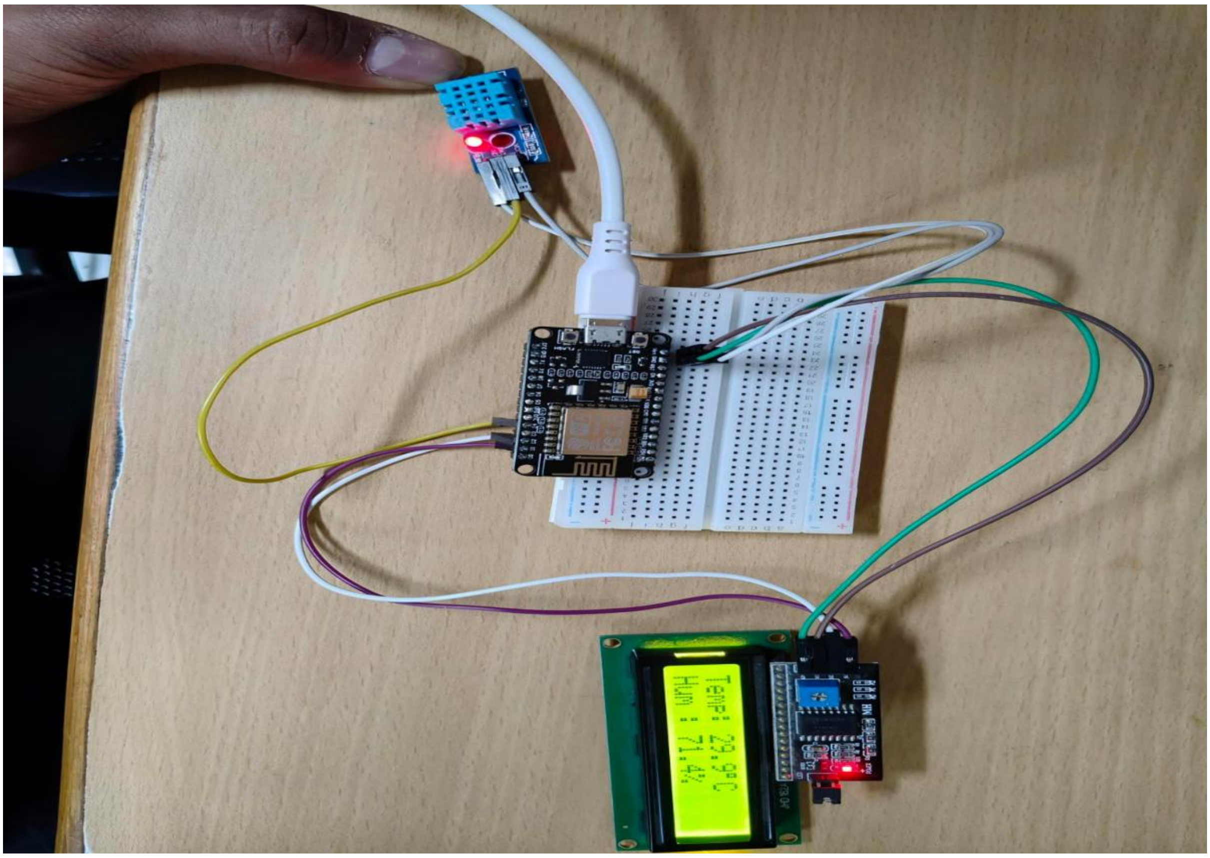
A 16×2 LCD is integrated into the prototype to display the measured temperature and humidity values locally. A 1-channel relay module is connected to the ESP8266 to provide switching control for the HVAC system based on the programmed temperature limit.

All components are assembled on a breadboard using jumper wires and appropriate power and ground connections. The ESP8266 is powered through USB, while the sensor, LCD, and relay are connected to the required supply lines.

The ESP8266 is programmed using the Arduino IDE. The program reads the DHT22 values, displays them on the LCD, checks the temperature condition, and controls the relay accordingly. The controller also establishes a Wi-Fi connection and uses MQTT to transmit the sensor readings to ThingSpeak Cloud.

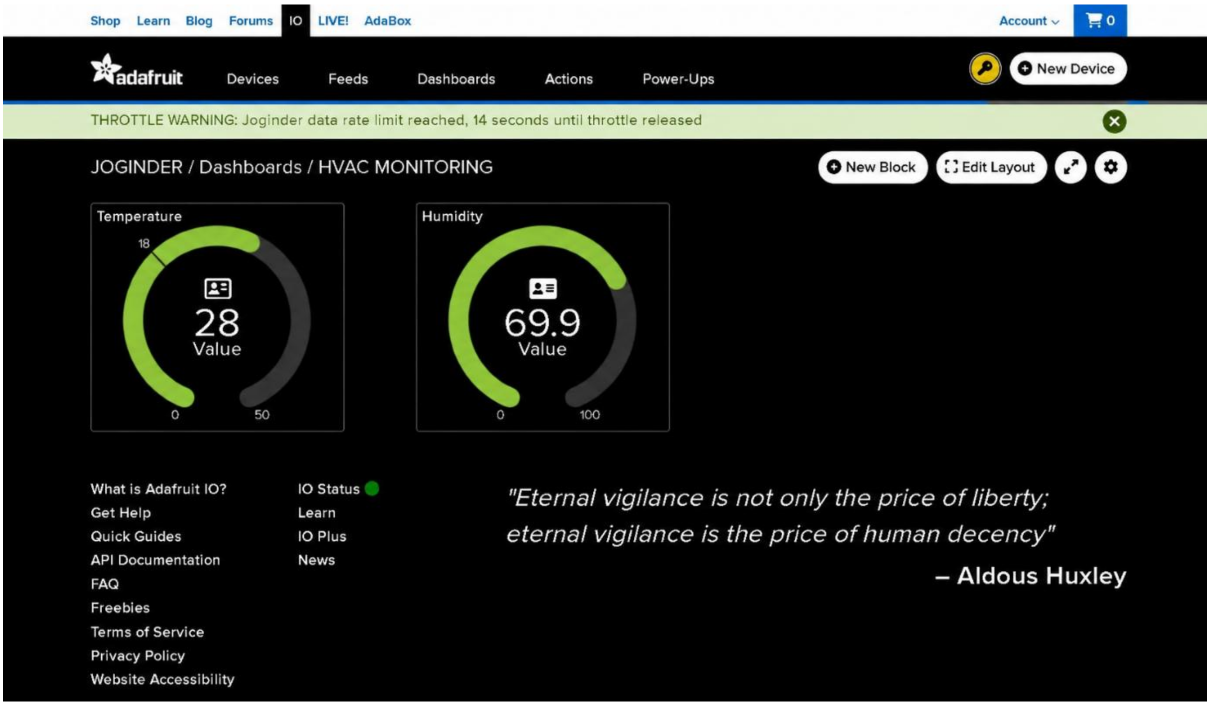
The ThingSpeak platform is configured to receive and store temperature and humidity data. The cloud dashboard provides graphical visualization, allowing the system performance and environmental changes to be monitored remotely.

The implemented prototype therefore combines sensor measurement, local display, automatic HVAC control, Wi-Fi connectivity, MQTT communication, and ThingSpeak Cloud monitoring in a single working system



RESULT & PERFORMANCE ANALYSIS

The developed MQTT-Based Remote HVAC Monitoring System was successfully implemented using the ESP8266 NodeMCU, DHT22 sensor, 16×2 LCD, relay module, MQTT communication, and ThingSpeak Cloud. The prototype demonstrated successful sensing, wireless communication, cloud monitoring, and HVAC control.



RESULT

- The DHT22 continuously measured temperature and humidity.
- The 16×2 LCD displayed real-time sensor readings.
- ESP8266 successfully established Wi-Fi connectivity.
- MQTT enabled efficient transmission of sensor data.
- ThingSpeak stored and visualized the collected data.
- The relay provided HVAC ON/OFF control based on the programmed temperature limit.
- Remote monitoring was achieved through the ThingSpeak dashboard.
- Historical temperature and humidity data available for analysis.
- The system operated continuously with repeated sensing and data transmission.